

Problem Set 5
Intermediate Microeconomics
Fall 2022

1. Luke maximizes his utility over leisure (N) and income (Y) given the following utility function:

$$U(Y, N) = Y + \ln N,$$

subject to the following constraints:

$$H = 24 - N \text{ and}$$

$$Y = wH + Y^*,$$

where H is the hours of work per day, w is the wage rate, and Y^* is her unearned income.

- Derive Luke's labor supply function.
- How does an increase in wage rate, w , affect his labor supply and income (holding other things equal)? Can you say anything about the relative magnitude of the substitution and income effects of a wage increase?
- How does an increase in unearned income, Y^* , affect his labor supply and income?
- Now, the government imposes a tax rate, τ , on her wage income. Write down his new budget constraint.
- How does an increase in τ affect his labor supply and income?

a. $\frac{MU_N}{MU_Y} = w \Rightarrow \frac{1/N}{1} = w \Rightarrow \frac{1}{N} = w \Rightarrow \frac{1}{w} = N^*$

$$Y = w(24 - N) + Y^* \Rightarrow wN + Y = 24w + Y^* \Rightarrow \frac{1}{N} \cdot N + Y = 24w + Y^* \Rightarrow Y = 24w + Y^* - 1$$

$$H^* = 24 - N = 24 - \frac{1}{w}$$

b. $\frac{\partial H^*}{\partial w} = -\frac{0-1}{w^2} = \frac{1}{w^2} > 0$ $\frac{\partial Y}{\partial w} = 24 > 0$

There is not IE for the $U = \ln N + Y$, so $|SE| > |IE| = 0$

c. $\frac{\partial H}{\partial Y^*} = 0$ $\frac{\partial Y}{\partial Y^*} = 1 > 0$ no affect on labor supply

d. $\max U = \ln N + Y$

s.t. $H = 24 - N$, $Y = wH + Y^*$

$$Y = (1-\tau)w(24 - N) + Y^* \Rightarrow Y = 24w(1-\tau) + Y^* - 1$$

$$H = 24 - \frac{1}{(1-\tau)w} = 24 - \frac{1}{w-\tau w}$$

e. $\frac{\partial H}{\partial \tau} = -\frac{1}{(1-\tau)^2 w} < 0$ $\frac{\partial Y}{\partial \tau} = -24w < 0$

$$= -\frac{1}{w-\tau w} - \frac{+w}{(w-\tau w)^2} = -\frac{w}{(1-\tau)^2 w^2} = -\frac{1}{(1-\tau)^2 w} < 0$$

*4.18 The government collects a specific tax of t for each hour worked. Thus, a worker whose wage is w keeps $w - t$ after taxes and supplies $H(w - t)$ hours of work. The government wants to know if its tax revenue will increase or decrease if it lowers t . Show that how the tax revenue changes depends on the elasticity of supply of labor, η . (Hint: See Solved Problem 5.6.) **M**

after-tax wage $\underline{w} = w - t$ $T = t H(w - t) = t H(\underline{w})$



$$\eta = \frac{dH(\underline{w})}{d\underline{w}} \times \frac{\underline{w}}{H(\underline{w})}$$

$$\frac{\partial T}{\partial t} = H(\underline{w}) + t \frac{dH}{d\underline{w}} \times \frac{d\underline{w}}{dt}$$

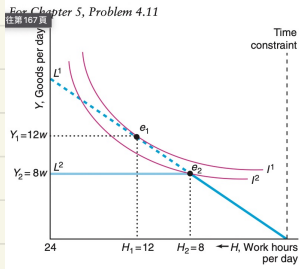
$$= H(\underline{w}) + t \frac{dH}{d\underline{w}} \times (-1) = H(\underline{w}) - t \frac{dH}{d\underline{w}}$$

$$\frac{\partial T}{\partial t} \begin{matrix} \uparrow \\ \downarrow \end{matrix} < 0 \Rightarrow H(\underline{w}) - t \frac{dH}{d\underline{w}} < 0$$

$$H(\underline{w}) < t \frac{dH}{d\underline{w}} \Rightarrow \frac{1}{t} < \frac{dH}{d\underline{w}} \times \frac{1}{H(\underline{w})}$$

$$\frac{w-t}{t} < \frac{dH}{d\underline{w}} \times \frac{\underline{w}}{H(\underline{w})} \Rightarrow \frac{w}{t} < \eta$$

- *4.11 Originally, Julia could work as many hours as she wanted at a wage of w . She chose to work 12 hours per day. Then, her employer told her that, in the future, she may work as many hours as she wants up to a maximum of 8 hours (and she can find no additional part-time job). How does her optimal choice between leisure and goods change? Does this change hurt her?



Original L_1 with a slope $-w$

$$Y_1 = 12w, I_1$$

New L_2 with maximum-hours restriction

$$Y_2 = 8w, I_2$$

I_2 must be below I_1

$$a. \max_N Y + \ln N$$

$$s.t. \quad wN + Y = 2q w + Y^*$$

$$\begin{cases} \frac{1}{N} = w \cdot 1 \\ \end{cases}$$

$$\begin{cases} wN + Y = 2q w + Y^* \end{cases}$$

$$\Rightarrow N = \frac{1}{w} \Rightarrow H = 2q - \frac{1}{w}, \quad Y = 2q w + Y^* - 1$$

$$b. \quad \frac{\partial H}{\partial w} = \frac{1}{w^2} > 0, \quad \frac{\partial Y}{\partial w} = 2q > 0$$

There is no income effect for the quasi-linear utility fn., so $|SE| > |IE| = 0$.

$$c. \quad \frac{\partial H}{\partial Y^*} = 0 \quad (\text{no IE}), \quad \frac{\partial Y}{\partial Y^*} = 1 > 0$$

$$d. \quad (1-z)wN + Y = 2q(1-z)w + Y^*$$

$$e. \quad \max_N Y + \ln N \quad s.t. \quad (1-z)wN + Y = 2q(1-z)w + Y^*$$

$$\frac{1}{N} = (1-z)w \Rightarrow N = \frac{1}{(1-z)w} \Rightarrow H = 2q - \frac{1}{(1-z)w}$$

$$\Rightarrow Y = 2q(1-z)w + Y^* - 1$$

$$\Rightarrow \frac{\partial H}{\partial z} = -\frac{1}{(1-z)^2 w} < 0, \quad \frac{\partial Y}{\partial z} = -2q w < 0$$

2. In class, we introduce a two-period consumption model. Suppose now the government imposes a lump-sum tax T_1 in period 1 and T_2 in period 2. The consumer's optimization problem is given as follows:

$$\begin{aligned} \max \quad & u(c_1) + \beta u(c_2) \\ \text{s.t.} \quad & \begin{cases} c_1 + b_1 = y_1 - T_1, \\ c_2 = y_2 - T_2 + (1+r)b_1. \end{cases} \end{aligned}$$

- a. What are the marginal cost and marginal benefits of saving? **Use your intuition** to derive the first order condition for optimal consumption in the two periods.

We will solve this problem in class on January 4.

$$mC : \beta u'(c_2) = \beta mu_2$$

$$mB : u'(c_1) = mu_1$$

- b. Suppose the government decreases the lump-sum tax in the first period by G and raise the lump-sum tax in the second period by $(1+r)G$. Graphically discuss the effect of this temporary tax cut on his consumption in the two periods and his saving. Use $u(c) = \ln C$ to verify your answer.

We will solve this problem in class on January 4.